

Engineering Chemistry

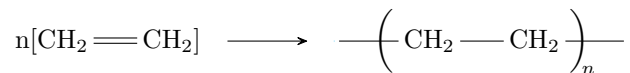
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April 22, 2025

1 Polymers

1.1 Introduction:

1. Polymers are made up of n number of "monomer units" (atomic weight less than 1000)
2. Atomic weight of polymers is more than 30,000.
3. Monomer $\xrightarrow{\text{Initiator}}$ Polymer (Polymerization)
4. A true monomer (a monomer from which a polymer can be produced) must have two bonding sites or two functional groups.
5. A double bond is a single functional group; but it has two bonding sites.
6. Example:



1.2 Methods of Polymerization:

1.2.1 Bulk Polymerization:

1. Here, the monomer is in liquid state, and no additional solvent is used (the monomer itself acts as the solvent).
2. The monomer is converted to polymer by mixing while heating.
3. This is an exothermic reaction, and we don't have any control over the temperature of the reaction. Hence, explosion could occur.
4. **Advantages:**
 - (a) Product is 100% formed, no monomer is wasted.
 - (b) Purification is easy as no additional solvent is used.
5. **Disadvantages:**

- (a) Limited to only liquid monomers.
- (b) Viscosity builds up over time, which cause difficulty in mixing.
- (c) Heat control is difficult.
- (d) Broad distribution of molecular mass.
- (e) **Mass Control:** Monomer is stuck and cannot reach the place of reaction.
- (f) **Gel Effect:** Auto-acceleration of reaction.

6. **Examples:** Plexiglass, Polystyrene.

1.2.2 Solution Polymerization:

1. The monomer (solute) is in any state, and it is dissolved in an appropriate inert solvent.
2. **Advantages:**
 - (a) Proper mixing takes place due to addition of external inert solvent.
 - (b) Viscosity does not build up.
 - (c) Mass and Heat Transfer are easy.
 - (d) If solvent is not required to be removed, these polymers can be used as adhesives.
3. **Advantages:**
 - (a) Solvent may as a terminating agent, and can end the reaction prematurely.
 - (b) As removal of last trace of solvent may be difficult, purification of the polymer is also difficult.
 - (c) Limitations of molecular weight.
4. **Examples:** Polyvinylchloride, Polyethylene.

1.2.3 Suspension Polymerization:

1. The monomer is a water-insoluble solute. Water is always the solvent, and the initiator is soluble in water.
2. To avoid coagulation, stabilizing and anti-coagulating agents are added.
3. This is also called as pearl or bead polymerization.
4. **Advantages:**
 - (a) Water is used as a solvent - which is economical, cheap, and non-toxic.
 - (b) Purification is easy and can be done by simple filtration.

5. **Disadvantages:**

- (a) This method can only be used for water in-soluble monomers.
- (b) Water is the only solvent used here.
- (c) The molecular weight control is difficult.

6. **Examples:** Polyvinylacetate, styrene-divinylbenzene.

1.2.4 Emulsion Polymerization:

1. Emulsifying Agent: R-COONa / R-SO₃Na.
2. The monomer is a water-insoluble solute. Water is always the solvent, and the initiator is soluble in water.
3. The monomer is hydrophobic, and the initiator is hydrophilic.
4. Once the emulsifying agent is added to the water, it forms a micelle at CMC and then the monomers goes inside the micelle.

5. **Advantages:**

- (a) Water is used as a solvent - which is economical, cheap, and non-toxic.
- (b) We can synthesize commercial, large-size polymers.
- (c) Elastomers can be synthesized.

6. **Disadvantages:**

- (a) This method can only be used for water in-soluble monomers.
- (b) Water is the only solvent used here.
- (c) Purification is very difficult.

7. **Examples:** Polyvinylacetate, Polyvinylchloride.

1.3 Determination of molecular weight of polymers:

1. **Degree of Polymerization:** The number of monomer units in a polymer. Molecular weight of polymer = DP × Molecular weight of monomer.

2. **Number Average Molecular Weight:**

$$M_n = \frac{\sum N_i M_i}{\sum N_i} = \frac{N_1 M_1 + N_2 M_2 + \dots}{N_1 + N_2 + \dots}$$

3. **Weight Average Molecular Weight:**

$$M_w = \frac{\sum N_i W_i^2}{\sum N_i W_i} = \frac{N_1 W_1^2 + N_2 W_2^2 + \dots}{N_1 W_1 + N_2 W_2 + \dots}$$

4. **Polydispersity Index:** It is the measure of heterogeneity of the polymer.

$$PDI = \frac{M_w}{M_n}$$

It is always greater than 1.

5. **Question:** For the given data, find M_n , M_w , and PDI.

$$N_1 = 10, N_2 = 5, M_1 = 10000, M_2 = 100000$$

Solution:

$$M_n = \frac{10 \times 10000 + 5 \times 100000}{10 + 5} = 40000$$

$$M_w = \frac{10 \times 10000^2 + 5 \times 100000^2}{10 \times 10000 + 5 \times 100000} = 85000$$

$$PDI = \frac{85000}{40000} = 2.125$$

1.3.1 Instruments to determine average molecular weights:

Number Average Molecular Weight:

1. Gel Permeation Chromatography (GPC).
2. Osmometry.

Weight Average Molecular Weight:

1. Small Angle Light Scattering.
2. Small Angle Neutron Scattering.
3. X-Ray Scattering.

1.4 Classification of Polymers Upon Nature of Monomers:

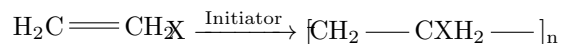
1. **Homopolymers:** Polymers made up of a single type of monomer. (Example: Polyethylene)
2. **Copolymers:** Polymers made up of two different types of monomers. (Example: Nylon-6,6)
3. **Types of Copolymers:**
 - (a) **Alternating Copolymers:** Monomers are arranged in an alternate fashion. (Example: Nylon-6,6)
 - (b) **Random Copolymers:** Monomers are arranged in a random fashion. (Example: Styrene-Butadiene Rubber (SBR))
 - (c) **Block Copolymers:** Monomers are arranged in blocks. (Example: Polystyrene-b-Polymethylmethacrylate)
 - (d) **Graft Copolymers:** One monomer is the backbone, and the other monomer is grafted on it. (Example: Polyethylene-g-Poly(n-butylacrylate))

1.5 Classification of Polymers Upon Synthesis:

1.5.1 Addition Polymerization:

1. This is also called as "Chain Polymerization".

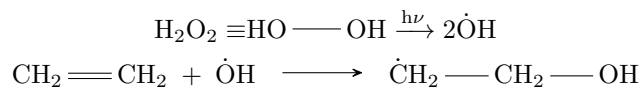
2.



3. The monomer has a olefilic bond.
4. No elimination of molecules occur.
5. Elemental composition of monomer and polymer is same.

6. Free Radical Mechanism:

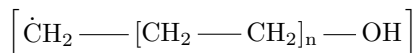
(a) Initiation:



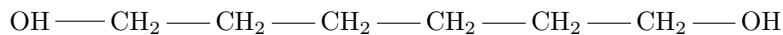
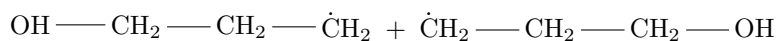
(b) Propagation:



and so on, forming



(c) Termination:

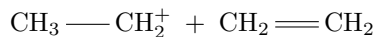


7. Cationic Mechanism:

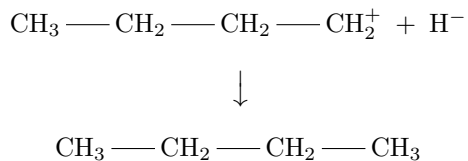
(a) Initiation: The initiator is a lewis acid / electrophile.



(b) Propagation:

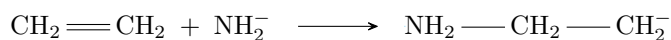


(c) Termination: The terminator is a lewis base / nucleophile.

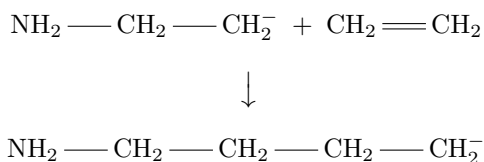


8. Anionic Mechanism:

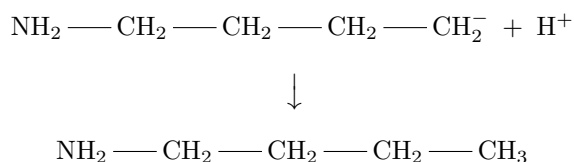
(a) Initiation: The initiator is a lewis base / nucleophile.



(b) Propagation:

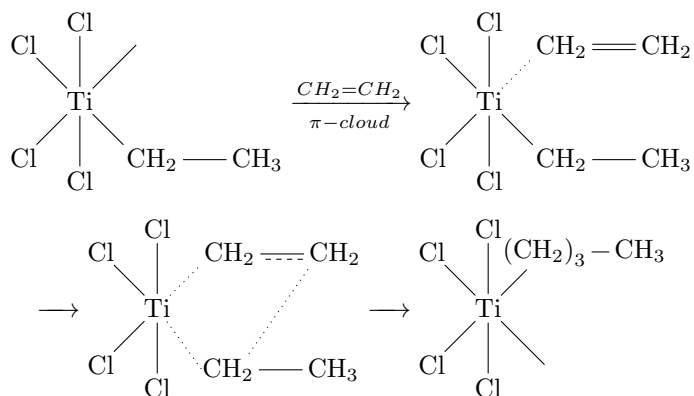


(c) Termination: The terminator is a lewis acid / electrophile.



9. Ziegler-Natta Mechanism:

Catalyst: Ziegler-Natta Catalyst. ($\text{TiCl}_4 + \text{Al}(\text{C}_2\text{H}_5)_3$).



10. Advantages:

(a) Linear polymer chains can be synthesized.

(b) Stereoregular polymers can be synthesized.

11.

Low Density Polyethylene	High Density Polyethylene
Radical Polymerization + Peroxide	Ziegler-Natta Mechanism
Density = 0.92gm/cm ³	Density = 0.97gm/cm ³
Temperature = 300-450°C	Temperature = 250-300°C
Pressure = 1000-2000 atm	Pressure = 6-7 atm
Less Mechanical Strength	More Mechanical Strength

12. **Tacticity:** The arrangement of functional groups on the main chain.

- (a) Isotactic: All functional groups are on the same side. (Example: Polypropylene).
- (b) Syndiotactic: Alternating arrangement of functional groups. (Example: Gutla-Percha).
- (c) Atactic: Random arrangement of functional groups. (Example: Polystyrene).

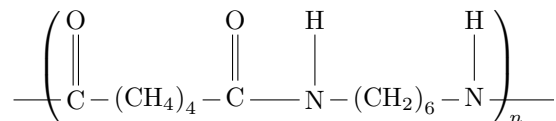
1.5.2 Condensation Polymerization:

1. Polyamides:

(a) It has an amide group.

(b) **Nylon-6,6:**

- i. Monomers: Adipic Acid and Hexamethylene Diamine.
- ii.

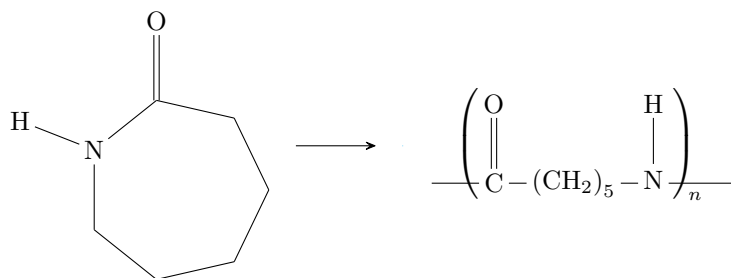


- iii. It shows extended intermolecular hydrogen bonding.
- iv. It is a copolymer.
- v. It is a thermoplastic.
- vi. Nylon 6,6 is used for airbags, tires, ropes, conveyor belts, etc.
- vii. It is a light material so it is suitable to be used for parachutes.

(c) **Nylon-6:**

- i. Monomers: Caprolactum.

ii.



- iii. It shows intramolecular hydrogen bonding.
- iv. It is a homopolymer.
- v. It is a thermoplastic.
- vi. Nylon 6 is widely used in clothing, sportswear, and outdoor gear due to its strength and elasticity.
- vii. It is used in the automotive industry for gears and bearings.

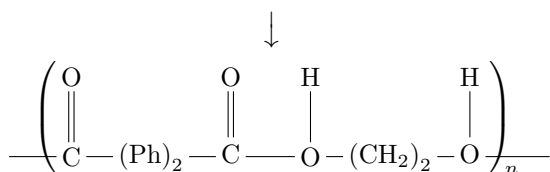
2. Polyesters:

(a) It has an ester group.

(b) **Polyethylene Terephthalate (PET):**

i. Monomers: Ethylene Glycol and Terephthalic Acid.

ii.



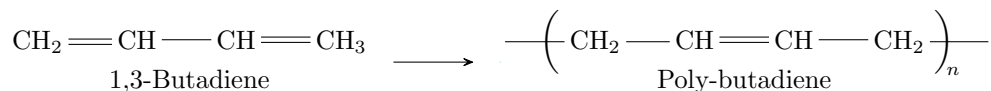
- iii. It is a copolymer.
- iv. It is a thermoplastic.
- v. PET is used in the production of bottles and food containers.
- vi. It is used in the textile industry for making clothes.
- vii. It is also called as "Dacron".

1.6 Classification Based Upon Intermolecular Forces:

1.6.1 Elastomers:

- 1. These have van-der waal forces.
- 2. Rubbers are elastomers.

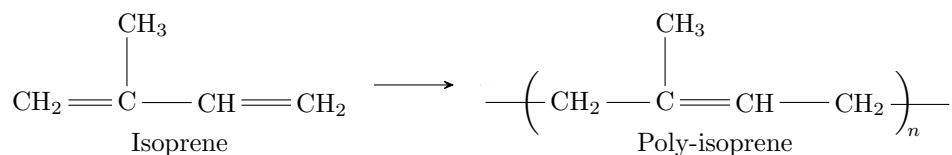
3. Polybutadiene:



It is used in the cores of golf balls to enhance their bounce.

Used in conveyor belts, shoe soles, and other rubber products.

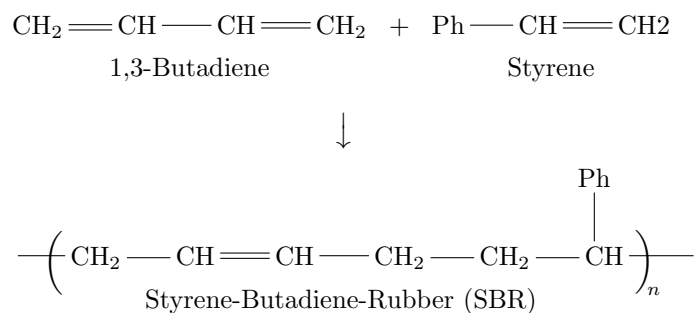
4. Polyisoprene:



It is used in adhesives for bonding materials like textiles.

It is used to make medical devices, including gloves and catheters.

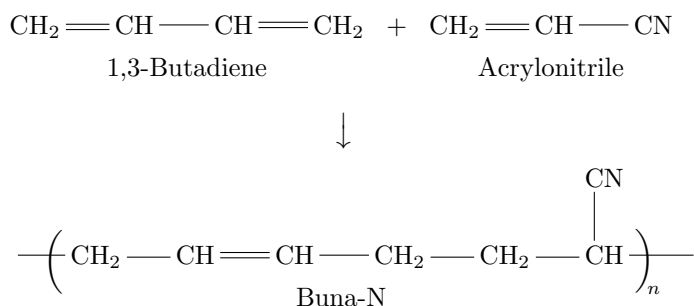
5. Buna-S:



It is commonly used in making shoe soles.

It is used in the production of conveyor belts.

6. Buna-N:



It is used in fuel and oil handling hoses.

It is commonly used in making seals and grommets.

1.6.2 Fibers:

1. They have extended intermolecular hydrogen bonding.
2. Examples: Nylon-6, Nylon-6,6.

1.6.3 Thermoplastics:

1. Moulding and Remoulding is possible as only physical changes occur on heating.
2. Examples: Polyethylene, Polyvinylchloride.

1.6.4 Thermosetting Plastics:

1. Cross-linking happen.
2. Remoulding is not possible as chemical bonds break on heating.
3. Examples: Bakelite.

1.7 Glass Transition Temperature (T_g):

1. The temperature at which the polymer changes from a glassy state to a rubbery state is called the "glass transition temperature (T_g)".
2. The temperature at which it melts is called the "flow temperature (T_f)".

1.7.1 Factors affecting T_g values:

1. More the crystallinity, more the T_g value.
2. More the molecular weight, more the T_g value.
3. More the intermolecular forces, more the T_g value.
4. More the flexibility, less the T_g value.
5. Bulky groups increase T_g value.

1.8 Some Commercial Polymers:

1.8.1 Polytetrafluoroethylene (PTFE):

1.

$\text{CF}_2 = \text{CF}_2$
Tetrafluoroethylene

$\longrightarrow$

$$\text{—}\left(\text{CF}_2\text{—CF}_2\right)_n\text{—}$$

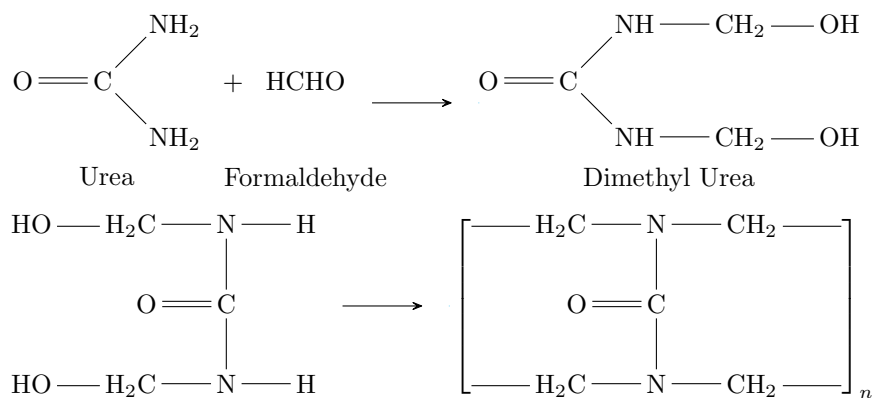
Teflon
2. This is formed through emulsion polymerization in presence of peroxide.
3. It is commercially called as "Teflon".

4. **Applications:**

- (a) It is used in non-stick cookware.
- (b) It is used as a lubricant to prevent captive insects from escaping.

1.8.2 **Urea Formaldehyde Resin (UF):**

1.



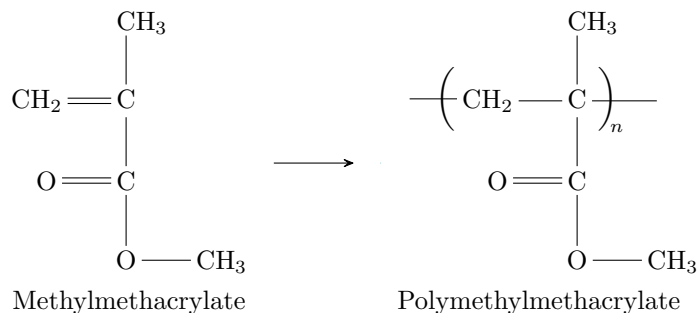
2. It is a thermosetting polymer and has cross-linking.

3. **Applications:**

- (a) It is used in making electrical switches.
- (b) It is used in making wrinkle resistant fabrics.

1.8.3 **Polymethylmethacrylate (PMMA):**

1.



2. It is commercially called as "Plexiglass".

3. **Applications:**

- (a) It is used to make windows of aircrafts.
- (b) It is used in making lenses of spectacles.

2 Water Technology

2.1 Introduction:

1. The technology used to improve the quality of water is called "water technology".
2. Water on earth majorly come from 3 sources: Oceans (97%), Glaciers and Underground Water (2.7%), and Freshwater (0.3%).
3. This freshwater also consists of impurities; mainly classified into three types: (i) Physical, (ii) Chemical, and (iii) Biological.
4. **Physical Impurities:**
 - (a) Turbidity: Presence of suspended particles (sand, dust, etc.)
 - (b) Odour: Dead organisms in water, compounds of N, S, P.
 - (c) Taste: Dissolution of excess minerals in water.
5. **Chemical Impurities:**
 - (a) Organic: Industrial Waste.
 - (b) Inorganic: Excess of minerals like CaHCO_3 , MgHCO_3 , etc.
 - (c) Dissolved Gases: CO_2 , NH_3 , O_2 , etc.
6. **Biological Impurities:** Microorganisms in water.

2.2 Hardness of Water:

1. The presence of Ca^{2+} and Mg^{2+} ions in water is called "hardness of water".
2. **Types of Hardness:**
 - (a) **Temporary Hardness:** It is due to the presence of bicarbonates of Ca and Mg.
 - (b) **Permanent Hardness:** It is due to the presence of chlorides and sulphates of Ca and Mg.

3.

$$\text{Hardness (Eq. of CaCO}_3) = W_{\text{Substance}} \cdot \frac{E_{\text{CaCO}_3}}{Eq_{\text{Substance}}}$$

2.2.1 Units of Hardness:

1. **mg/L:** 1 mg of substance per 1 litre of water.
2. **ppm:** 1 part of substance per 10^6 parts of water. ($\text{mg/L} = \text{ppm}$)
3. **Degree French ($^\circ F_r$):** 1 part of substance per 10^5 parts of water.
4. **Clarke's Degree ($^\circ Cl$):** 1 part of substance per 70000 parts of water.

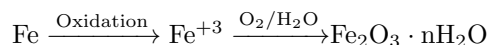
2.3 Disadvantages of Hard Water:

2.3.1 Sludge and Scales:

1. In steam generation boiler, when water with hardness is evaporated, the minerals build up in the boiler. This leads to the formation of:
 - (a) **Sludge:** Loose settlement of minerals in the boiler.
 - (b) **Scales:** Hard settlement of minerals in the boiler.
2. Continuously diluting the water in the boiler will reduce sludge and scales.
3. $MgCl_2, MgCO_3$ are soluble in hot water and form sludge in colder part of boiler.
4. $CaSO_4, CaCO_3$ settle down to form scales.
5. **Types of Boilers:**
 - (a) Low Pressure Boiler: Pressure of 300 - 600 psi.
 - (b) Medium Pressure Boiler: Pressure of 600 - 900 psi.
 - (c) High Pressure Boiler: Pressure of 900 - 1200 psi.
6. To remove scales, thermal shocks and chemical washing can be done.

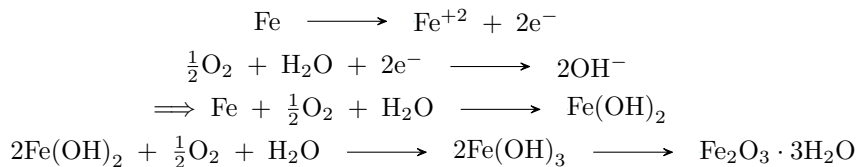
2.3.2 Boiler Corrosion:

Boiler is made up of iron metal (Fe):



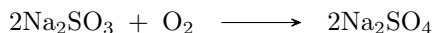
1. Corrosion due to O_2 :

(a) Reaction:

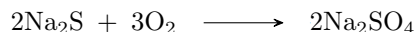


(b) Removal:

- i. By addition of Na_2SO_3 (sodium sulphite):



- ii. By addition of Na_2S (sodium sulphide):



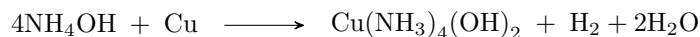
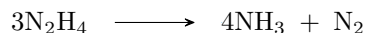
Problem: Formation of excess Na_2SO_4 can produce corrosive base and acid at high pressure:



iii. By addition of N_2H_4 (hydrazine):



Problem: Excess of N_2H_4 can decompose into NH_4OH which can corrode copper joints in the boiler:

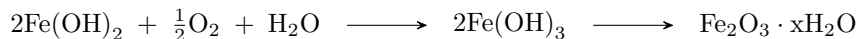
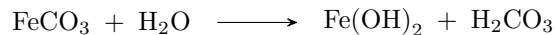
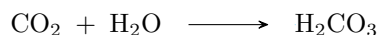


2. Corrosion due to CO_2 :

(a) **Source:**



(b) **Reaction:**

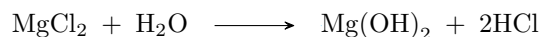


(c) **Removal:** Addition of NH_4OH forming a water soluble complex.

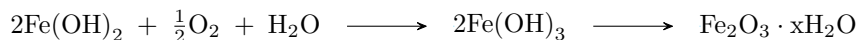


3. Corrosion due to Acids:

(a) **Source:** It is mainly due to the decomposition of salts yielding acids.



(b) **Reaction:**

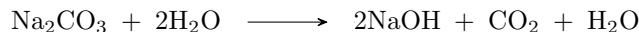


(c) **Removal:** Addition of quantitative amount of alkali.

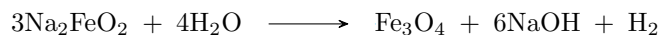
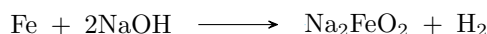
4. Caustic Embrittlement:

(a) It is the corrosion due to alkali.

(b) We use Na_2CO_3 excessively for treatment of water.



(c) **Reaction:**



2.4 Internal Water Treatment:

1. Here water is treated inside the boiler.
2. It is called as "corrective method".
3. It is mainly used in Low Pressure Boilers.

2.4.1 Colloidal Conditioning:

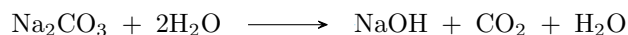
1. It forms a layer on the metal surface.
2. Example: Oil, Agar, Tanin, etc.

2.4.2 Carbonate Conditioning:

1. It is called as "sequestration", meaning one mineral gets converted into another form so it can be removed easily.
- 2.



3. If excess of Na_2CO_3 is added, it can lead to caustic embrittlement:



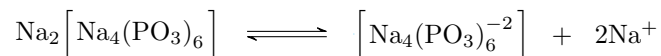
2.4.3 Phosphate Conditioning:



2.4.4 Calgon Conditioning:

1. Calgon is $\text{Na}_2[\text{Na}_4(\text{PO}_3)_6]$

- 2.

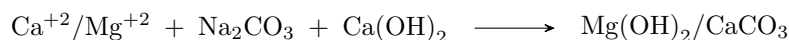


2.5 External Water Treatment:

1. Here water is treated outside the boiler.
2. It is called as "preventive method".
3. It is mainly used in High Pressure Boilers.

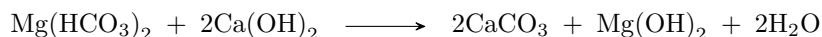
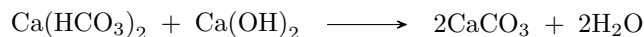
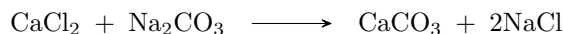
2.5.1 Soda Lime Process:

1. Na_2CO_3 and $Ca(OH)_2$ is used to soften the water.
- 2.

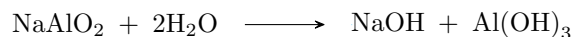


3. Cold Soda Lime Process:

(a)



- (b) Coagulant like $Al_2(OH)_3$ or $NaAlO_2$ is added to remove colloidal impurities.



- (c) Soft water obtained is of 50 - 60 ppm.

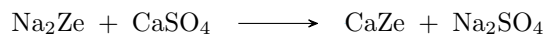
4. Hot Soda Lime Process:

- (a) At elevated temperature, reaction speed and efficiency increases.
- (b) Soft water obtained is of 15 - 30 ppm.
- (c) Coagulation is not required.

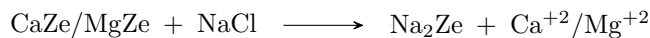
2.5.2 Zeolite Process:

1. Zeolites are $Na_2O \cdot Al_2O_3 \cdot xSiO_2 \cdot yH_2O$
2. They are called as "molecular sieves".
3. They are porous polymeric substances, which are classified into 2 types:
(a) Natural (very small pores) and (b) Synthetic (variable pore size).

4. Reactions:



5. Soft water obtained is of 10 - 15 ppm.
6. The obtained water is rich with sodium salt.
7. When all sodium ions are exchanged, the zeolite is saturated. To regenerate the zeolite, a 10% solution is added.

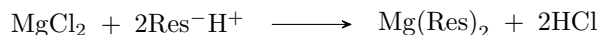
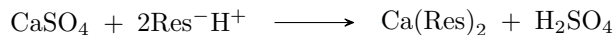


8. Turbidity has to be removed before treatment.
9. Water must be free from Mn^{+2} and Fe^{+2} before treatment.
10. Zeolites can be destroyed by acids.

2.5.3 Ion Exchange Process:

1. A long chain of cross-linked polymer resin, capable of exchanging ions in liquid form is used here.
2. $R \equiv -\text{SO}_3^- \text{H}^+ \implies$ Cation Exchange Resin ($\text{Res}^- \text{H}^+$).
3. $R \equiv -\text{CH}_2\text{N}^+(\text{CH}_3)_3\text{OH}^- \implies$ Anion Exchange Resin ($\text{Res}^- \text{OH}^+$).

4. At Cation Exchange Resin:

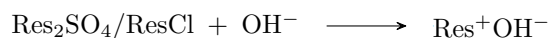
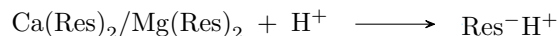


5. At Anion Exchange Resin:



6. Soft water obtained is of 2 - 3 ppm.

7. Regeneration:



2.6 Dissolved Oxygen (DO):

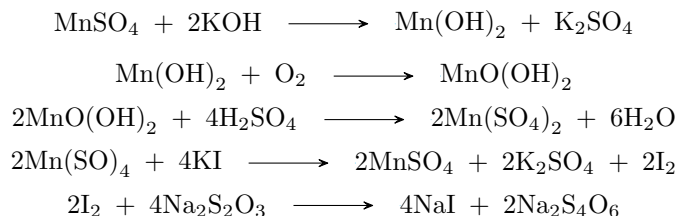
1. The amount of oxygen dissolved in water is called "dissolved oxygen".
2. It is measured in gm/l or ppm.
3. Generally, it is 0 - 18ppm
4. Natural water usually has 6ppm DO.

5. Factors Affecting DO:

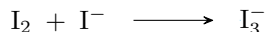
- (a) **Temperature:** More the temperature, less the DO.
- (b) **Altitude:** More the altitude, less the DO.
- (c) **Waste:** More the waste, less the DO.

2.6.1 Winkler's Method:

Take V ml of water with x ml of O₂ dissolved. Add MnSO₄, H₂SO₄, KOH, and KI to it. The reaction is:



Now,



This tri-iodide on reaction with starch forms a blue colored starch complex, which reacts with Na₂S₂O₃ to form a colorless solution. The amount of Na₂S₂O₃ used is proportional to the amount of O₂ dissolved.

4 moles of Na₂S₂O₃ $\equiv$ 1 mole of O₂ $\equiv$ 32gm of O₂.

$\therefore$ 1 mole of Na₂S₂O₃ $\equiv$ 8gm of O₂.

Hence, taking A ml of N moles of Na₂S₂O₃, we require V ml of water.

Hence, for 1L of water,

$$DO = \frac{8 \times A \times N \times 1000}{V} = \frac{8000 \times A \times N}{V}$$

2.7 Oxygen Demand:

1. The amount of oxygen required to oxidize the organic matter in water is called "oxygen demand".
2. **Types of Oxygen Demand:**
 - (a) Biological Oxygen Demand (BOD)
 - (b) Chemical Oxygen Demand (COD)

2.7.1 Biological Oxygen Demand (BOD):

1. It is the amount of oxygen required to oxidize biologically oxidizable impurities at 20°C for 5 days in water.
2. V ml of water is taken and is divided into 2 equal parts.
3. One part is processed by Winkler's method, called as "blank sample".
4. The other part is treated with bacteria (*Pseudomonas capica* / *Bacillus* sp1) in an incubator at 20°C for 5 days, and processed through Winkler's method.
5. The Dissolved Oxygen are DO_1 and DO_2 respectively.
- 6.

$$BOD = DO_1 - DO_2$$

$$BOD = \frac{8000 \times A \times N}{V} - \frac{8000 \times B \times N}{V} = \frac{8000 \times (A - B) \times N}{V}$$

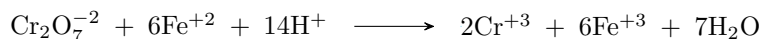
7. To make sure all oxygen doesn't get used up before 5 days, we dilute the sample.

$$\text{Dilution Factor, } p = \frac{V_1 \text{ (initial volume)}}{V_2 \text{ (final volume)}}$$

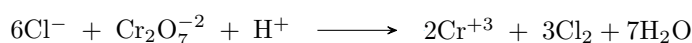
$$BOD = \frac{DO_1 - DO_2}{p} = \frac{8000 \times (A - B) \times N \times V_2}{V \times V_1}$$

2.7.2 Chemical Oxygen Demand (COD):

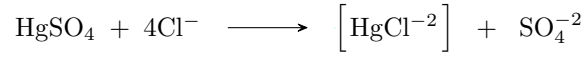
1. It is the amount of oxygen required to oxidize chemically oxidizable impurities using a O.A., Acid, and Catalyst.
2. $K_2Cr_2O_7 + H_2SO_4 + Ag_2SO_4 + HgSO_4$ is added to V ml of waste water and V ml of distilled water. The mixture is kept to reflux for 1.5 hours and some amount of $K_2Cr_2O_7$ is consumed.
3. After 1.5 hours, the solution is titrated with FAS in presence of ferroin indicator.
4. **Reaction:**



5. **Interference of Chlorine:**



6. To avoid interference, $HgSO_4$ is added:



7. As bacteria selectively oxidize organic matter, COD is always greater than BOD.

- 8.

$$COD = \frac{8000 \times (A - B) \times N}{V}$$

where A is amount of FAS in blank titration,

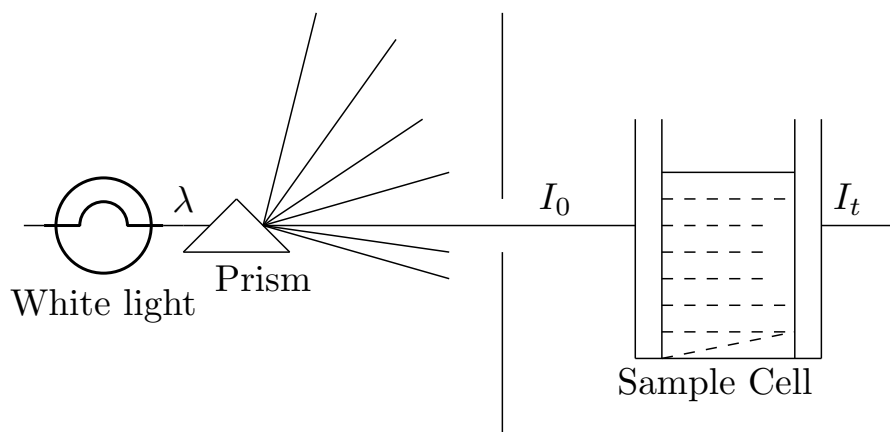
B is amount of FAS in waste water titration,

N is concentration of FAS, and V is volume of waste water.

3 Instrumental Methods of Chemical Analysis:

1. It is a collective of methods to characterise materials.
2. They are of 2 types:
 - (a) **Quantitative Methods:** Measures quantity of substance.
 - (b) **Qualitative Methods:** Confirms existence of substance.
3. Quantitative methods are of 2 types:
 - (a) **Volumetric Methods:** Using volume to measure quantity
 - (b) **Gramimetric Methods:** Using mass to measure quantity
4. Qualitative methods are of 2 types:
 - (a) **Electroanalytical Methods:** Involves measurement of voltage or current.
 - (b) **Spectrochemical Methods:** Involves EM Radiations with substance.

3.1 Colorimetry:



1. Substance absorbs monochromatic light from visible spectrum to go to an excited state ($M + h\nu \rightarrow M^*$)
2. It comes back to the ground state and releases the photon as heat.
3. Transmittance (T) is the fraction of specific wavelength of light that passes through the absorbing material.

$$T = \frac{I_t}{I_0}$$

4. Absorbance (A) is the fraction of specific wavelength of light absorbed by the absorbing material.

$$A = -\log(T) = \log\left(\frac{I_0}{I_t}\right)$$

5. In reality the real transmittance is always greater than the calculated transmittance as loss due to scattering and reflection is not considered in the calculations.
6. To fix this, the experiment is performed two times:
- (i) It is calculated only with solvent ($I_{solvent} \approx I_0$)
 - (ii) It is calculated of the entire solution ($I_{solution} \approx I_t$)

$$T = \frac{I_t}{I_0} = \frac{I_{solution}}{I_{solvent}}$$

7. **Lambert's Law:** The decrease of intensity of light with respect to the thickness of the sample is proportional to intensity of incident light.

$$-\frac{dI}{dl} = kI \implies \ln\left|\frac{I_0}{I_t}\right| = kl \implies \log\left(\frac{I_0}{I_t}\right) = \frac{k}{2.303}l$$

This can be written as: $A = \varepsilon l$

where ε is the molar absorptivity of the substance.

For a given wavelength λ , ε is constant.

8. **Beer-Lambert's Law:** It is an extension of Lambert's Law, and it states that:

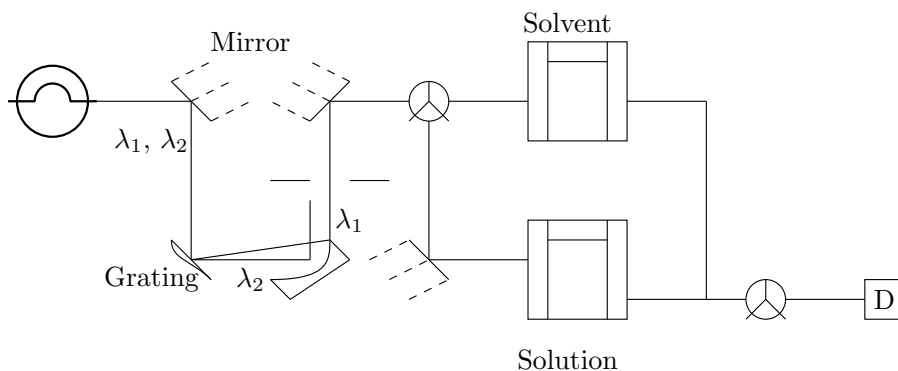
$$-\frac{dI}{dl} = kCI \implies A = \varepsilon Cl$$

where C is the concentration of the substance in the solution and l is the length of the sample cell.

Hence, as for a given wavelength λ , ε is constant, meaning $A \propto C$

9. **Application:** It is used in the estimation of concentration of a substance in a solution.
10. **Limitations:**
- (a) It only works for low concentrated solutions; the linearity is lost at higher concentration due to Brownian Movement.
 - (b) It only works for monochromatic light; hence instrumental errors can occur for various wavelengths.
 - (c) Chemical Decomposition may occur as incident light can break bonds.

11. Double Beam Spectrometer:



Assume that λ_1 and λ_2 are the two wavelengths of light entering the spectrometer.

At λ_1 :

$$A' = -\log(T') = \log\left(\frac{I'_0}{I'_t}\right)$$

$$\therefore A' = \varepsilon'Cl = \log\left(\frac{I'_0}{I'_t}\right) \Rightarrow I'_t = I'_0 \cdot 10^{-\varepsilon'Cl}$$

Similarly, **At λ_2 :**

$$I''_t = I''_0 \cdot 10^{-\varepsilon''Cl}$$

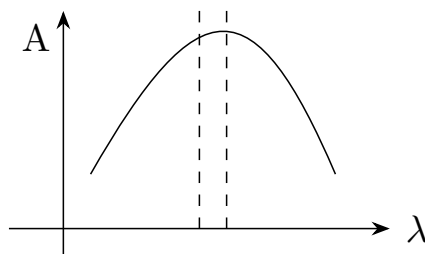
Now, net absorbance:

$$A = \log\left(\frac{I'_0 + I''_0}{I'_t + I''_t}\right) = \log\left(\frac{I'_0 + I''_0}{I'_0 \cdot 10^{-\varepsilon'Cl} + I''_0 \cdot 10^{-\varepsilon''Cl}}\right)$$

If $\varepsilon_1 \neq \varepsilon_2$, then it deviates from linearity.

If $\varepsilon_1 = \varepsilon_2 = \varepsilon$, then it is linear and $A = \varepsilon Cl$.

12. If C and l are constant, then:



13. **Question:** If $\lambda = 532nm$, $C = 0.8\mu m$, $T = 44\%$, $l = 2cm$, then find ε

Solution:

$$A = -\log(T) = -\log(0.44) = 0.356$$

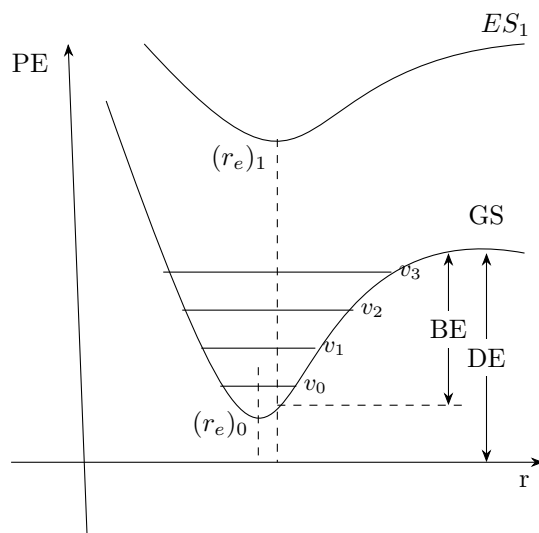
$$A = \varepsilon Cl \Rightarrow \varepsilon = \frac{A}{Cl} = \frac{0.356}{(0.8 \times 10^{-6})(2 \times 10^{-2})} = 2.225 \times 10^7 L/mol/m$$

3.2 UV-Visible Spectroscopy:

It is similar to colorimetry.

3.2.1 Morse Curve:

1. It is the plot of the potential energy of a molecule against the intermolecular distance of its constituent atoms.
2. For example, in H_2 ,



3. At room temperature, most molecules are at v_0 vibration state. This is called as Zero Point Energy State.
4. The energy required to separate the molecule from v_0 state is called as the binding energy.

5. The depth of the curve is directly proportional to the strength of the bond.
6. The width of the curve is directly proportional to the amplitude of vibration.
7. The more the slope of the curve, the more stiff the bond is. The less the slope of the curve the more soft the bond is.
8. $\Delta E_e \gg \Delta E_v \gg \Delta E_r$

3.2.2 Selection Rules:

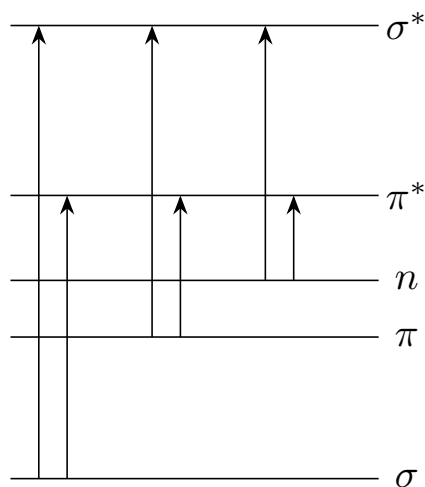
1. **Frank-Condon Principle:** The transition between two states is instantaneous. Hence, the molecule has the same intermolecular distance.
2. **Spin Selection Rule:** There is no selection for an electron to move from one vibrational state to another. However there is selection to move from one energy state to another

$$\Delta S = 0$$

3. **Orbital Selection Rule:** The transition is allowed only for $\Delta l = \pm 1$.
4. **Laporte Selection Rule:** For a molecule with center of inversion, the transition is allowed only for $g \leftrightarrow u$. It is forbidden for $g \leftrightarrow g$ and $u \leftrightarrow u$.

3.2.3 Origin of Electron Transitions:

1. Look at the given transition diagram.



2. Due to energy limitations, $\sigma - \pi^*$, $\sigma - \sigma^*$, and $\pi - \sigma^*$ are not possible.

3. $n - \sigma^*$ **Transition:**

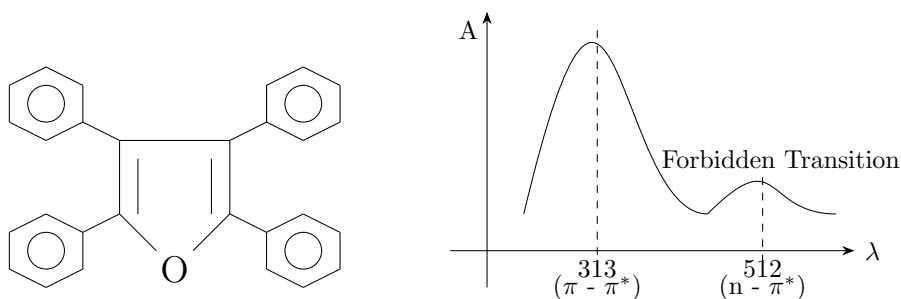
- (a) Depending upon the energy requirement, this transition may or may not be seen.
- (b) For this transition, the molecule must contain a functional group having non bonding and empty σ^* orbital.
- (c) Example: H_2O (167nm), CH_3Cl (169nm)

4. $n - \pi^*$ **Transition:**

- (a) For this transition, the molecule must contain a functional group having non bonding and empty π^* orbital.
- (b) Example: $(CH_3)_2CO$ (264nm in water, 279nm in hexane)

5. $\pi - \pi^*$ **Transition:**

- (a) For this transition, the molecule must contain an electron in a π orbital an empty π^* orbital.
- (b) Example:

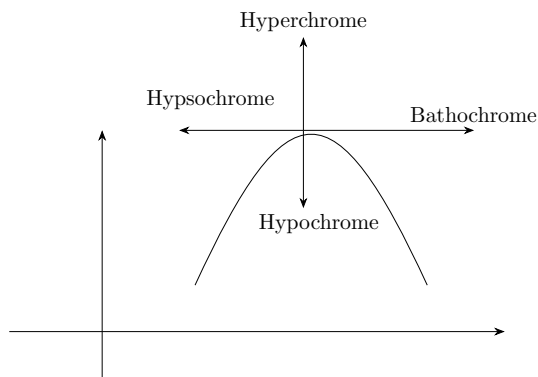


6. **Chromophore:** They are functional groups which give colour to the compound. They absorb electromagnetic radiations in the UV-Visible region.

7. **Auxochromes:** They are functional groups which do not absorb electromagnetic radiations in the UV-Visible region but have an influence on the chromophore. They may change the λ_{max} as well as the ϵ values of the molecule.

- (a) **Bathochromic Shift:** The shift of λ_{max} to the red region (longer wavelengths).

- (b) **Hypsochromic Shift:** The shift of λ_{max} to the blue region (shorter wavelengths).
- (c) **Hyperchromic Shift:** The increase in ϵ values.
- (d) **Hypochromic Shift:** The decrease in ϵ values.
- (e) The following diagram shows the effect of shifts:

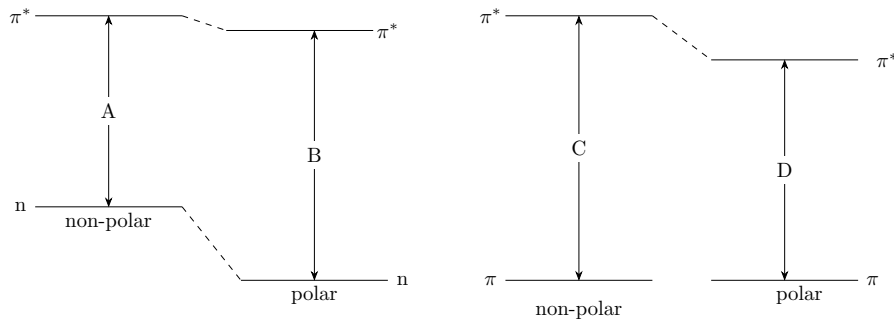


3.2.4 Factors Affecting UV-Visible Spectroscopy:

1. Solvent Requirements:

- (a) The solvent must have high purity.
- (b) It must not absorb radiation in the UV-Visible region.
- (c) It must have minimal interactions with the solute.
- (d) It must be cheap.

2. Effect of Polar Solvents:



- (a) In $n-\pi^*$ transitions, as $E_A < E_B$, hypsochromic shift is observed.
- (b) In $\pi-\pi^*$ transitions, as $E_C > E_D$, bathochromic shift is observed.

- 3. **Effect of Extended Conjugation:** The λ_{max} value increases with the increase in the number of conjugated double bonds.

3.2.5 Woodward-Fieser Rule (For di-enes):

1. Butadiene is taken as reference molecule ($\lambda_{max} = 217nm$).
2. For each conjugated double bond, the λ_{max} value increases by 30nm.
3. For double bond within the ring, the λ_{max} value increases by 36nm.
4. For exocyclic double bond, the λ_{max} value increases by 5nm.
5. For each methyl group, the λ_{max} value increases by 5nm.
6. For each ring residue, the λ_{max} value increases by 5nm.
7. For each O-alkyl group, the λ_{max} value increases by 6nm.
8. For each S-alkyl group, the λ_{max} value increases by 30nm.
9. For each N-alkyl group, the λ_{max} value increases by 60nm.
10. For each halogen atom, the λ_{max} value increases by 5nm.
11. For each O-acyl group, the λ_{max} value increases by 0nm.

3.2.6 Estimation of Concentration of Multiple Components:

For 2 components X and Y,

$$@ \lambda_1, A = \epsilon_X C_X l + \epsilon_Y C_Y l$$

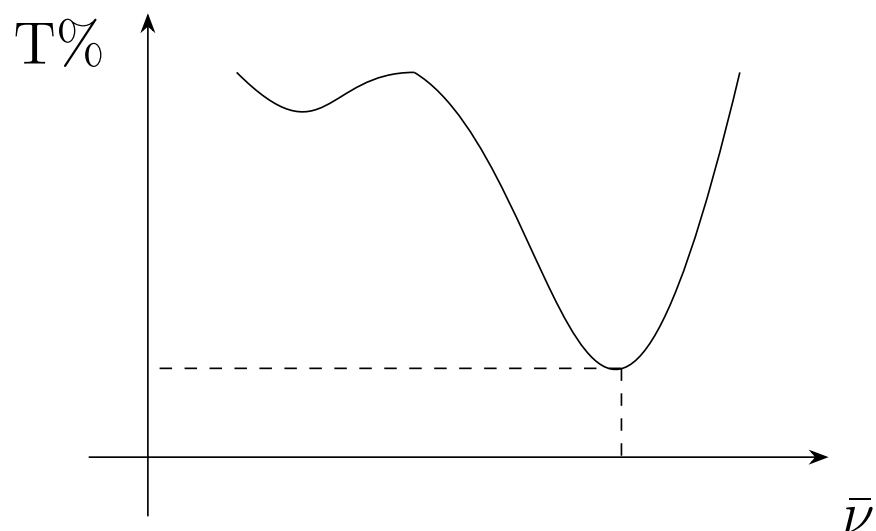
$$@ \lambda_2, A' = \epsilon'_X C_X l + \epsilon'_Y C_Y l$$

On solving the system, we get the C_X and C_Y .

For n components, n such equations are to be written.

3.3 Infrared Spectroscopy:

1. It is also called as "vibrational spectroscopy" as electrons go from one vibrational state to a higher state in the ground electronic state.
2. It is a plot between transmittance (T%) and wave number ($\bar{\nu}$).



3. For infrared spectroscopy the following conditions are to be satisfied:
 - (a) Vibrational frequency of the functional group should match with that of the EM radiation.
 - (b) Change in dipole moment must occur.
4. Infrared Spectroscopy is of 3 types:
 - (a) NIR (Near IR): $0.78 - 2.5\mu\text{m}$.
 - (b) MIR (Mid IR): $2.5 - 50\mu\text{m}$.
 - (c) FIR (Far IR): $50 - 1000\mu\text{m}$.

3.3.1 Vibrational Motion:

1. Vibrational Motion are of two types:
 - (a) Stretching Motion: It is the change in bond length.
 - i. Symmetric Stretching: The bond length of both the bonds increase or decrease simultaneously.
 - ii. Asymmetric Stretching: The bond length of one bond increases while the other decreases.
 - (b) Bending Motion: It is the change in bond angle.

- i. Scissoring: The bond angle decreases.
 - ii. Rocking: The bond angle increases.
 - iii. Wagging: The bond angle increases and decreases alternately.
 - iv. Twisting: The bond angle increases and decreases simultaneously.
2. Consider a spring-mass system with a spring constant k and mass m . The force acting on the system is given by:

$$ma = -kx \implies ma = F = -kx$$

Also, PE diagram is a parabolic curve as $PE = \frac{1}{2}kx^2$

$$\omega = \sqrt{\frac{k}{m}} \implies \nu = \frac{\omega}{2\pi} = \frac{1}{2\pi} \sqrt{\frac{k}{m}}$$

3. For two masses connected with a spring assuming a atom model,

$$\text{Reduced Mass, } \mu = \frac{m_1 m_2}{m_1 + m_2} \implies \nu = \frac{1}{2\pi} \sqrt{\frac{k}{\mu}} = \frac{1}{2\pi} \sqrt{\frac{k(m_1 + m_2)}{m_1 m_2}}$$

4. Energy of a vibrational state is given by: $E_v = (\frac{1}{2} + v)h\nu$ where "v" is called the "vibrational quantum number".

$$E_0 = \frac{1}{2}h\nu, \quad E_1 = \frac{3}{2}h\nu, \quad E_2 = \frac{5}{2}h\nu, \dots \implies \Delta E = h\nu$$

Now,

$$\nu = c\bar{\nu} \implies \bar{\nu} = \frac{\nu}{c} = \frac{1}{2\pi c} \sqrt{\frac{k(m_1 + m_2)}{m_1 m_2}}$$

5. As restoring force between atoms is weaker than spring mass system, the PE diagram does not follow the SHM parabolic path. Also uneven distribution of energy levels occur. Furthermore, as selection rules are also not followed, forbidden transitions can occur. These transitions are called as "overtones".
6. Values of k :
- (a) For single bond, $k = 0.5 \times 10^3 N/m$
 - (b) For double bond, $k = 1.0 \times 10^3 N/m$
 - (c) For triple bond, $k = 1.5 \times 10^3 N/m$

7. **For carbonyl group, -C=O**

$$m_1 = 12gm/mol = \frac{12 \times 10^{-3}}{6.023 \times 10^{23}} \approx 2kg$$

$$m_2 = 16 \text{ gm/mol} = \frac{16 \times 10^{-3}}{6.023 \times 10^{23}} \approx 2.6 \text{ kg}$$

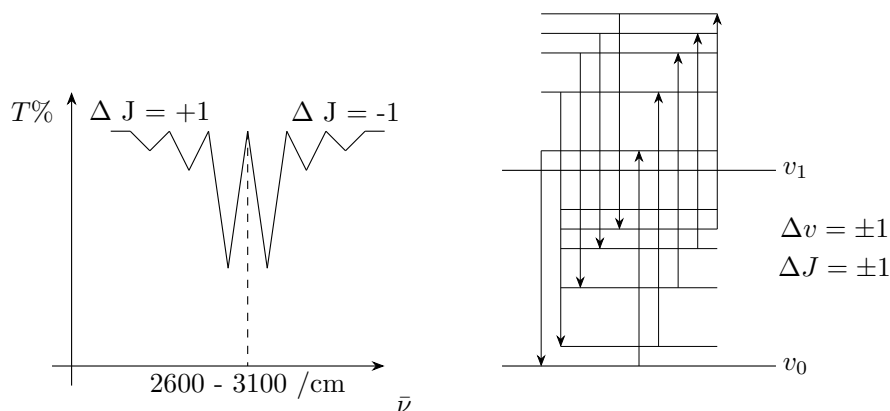
Reduced mass,

$$\mu = \frac{m_1 m_2}{m_1 + m_2} \approx \frac{2 \times 2.6}{2 + 2.6} \approx 1.1 \text{ kg}$$

$$\bar{\nu} = \frac{1}{2\pi c} \sqrt{\frac{k}{\mu}} = \frac{1}{2\pi(3 \times 10^8)} \sqrt{\frac{1 \times 10^3}{1.1}} = 1.6 \times 10^5 \text{ m}$$

Hence, for carbonyl group, $\bar{\nu} = 1600 \text{ cm}$

8. **Translational Motion:** Translation of the whole molecule along an axis.
9. **Rotational Motion:** Rotation of the whole molecule along an axis.
10. **Interatomic Motion:** Movement of an atom in the molecule wrt another.
11. Translational and rotational motions don't bring change in dipole moment; only interatomic motion brings change in dipole moment.
12. For a non-linear molecule, number of fundamental nodes / degrees of freedom (f) = 3N-6. (N is number of atoms in the molecule).
13. For a linear molecule, f = 3N-5.
14. For HCl, f = 3(2) - 5 = 1, meaning only one IR band must form (stretching). However, the spectroscopy shows the following result:

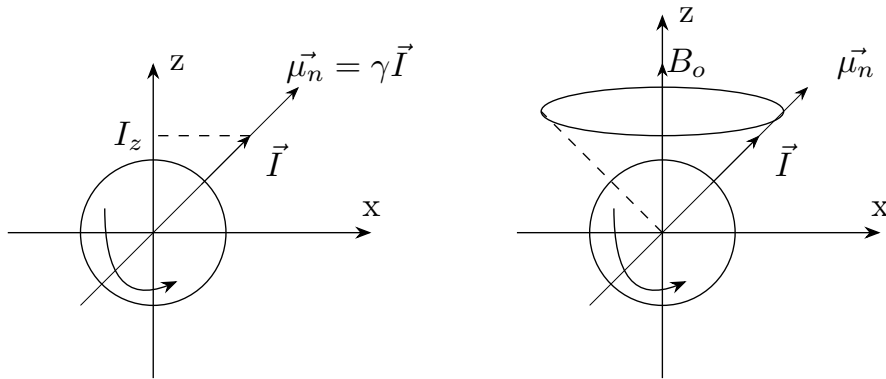


3.4 Magnetic Resonance Spectroscopy:

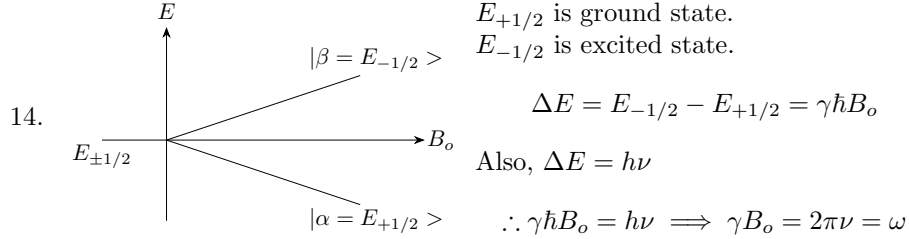
3.4.1 Nuclear Magnetic Resonance (NMR):

1. Here, the nucleus acts as a tiny bar magnet and 'I' is the nucleus spin.
2. It uses microwave (frequency much less than IR).

3. For atoms with odd mass number, I is a half integral.
(Ex., I for ${}^1_1H = \frac{1}{2}$, ${}^{11}_5B = \frac{3}{2}$, ${}^{13}_6C = \frac{1}{2}$)
4. For atoms with even mass number and odd atomic number, I is an integer.
(Ex., I for ${}^2_1H = 1$, ${}^{14}_7N = 1$)
5. For atoms with both even mass and atomic numbers, I is zero.
(Ex., I for ${}^4_2He = 0$, ${}^{12}_6C = 0$)
6. NMR for I = 0 is not possible.
7. Protons have an intrinsic property called as 'spin'. Due to this spin, an angular momentum arises.
8. $|\vec{I}| = \hbar\sqrt{I(I+1)}$ where I is the spin quantum number
For a proton, $I = \frac{1}{2} \implies |\vec{I}_p| = \frac{\hbar\sqrt{3}}{2}$
9. $I_z = \hbar m_I$ where m_I is magnetic quantum number.
The number of spin multiplicity = $2I+1 = 2. \implies m_I = \pm\frac{1}{2}$.

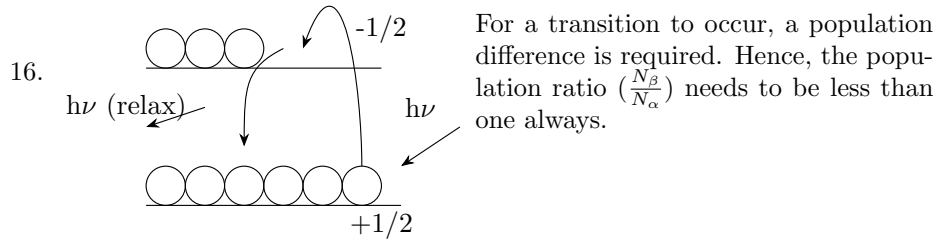


10. Nuclear Magnetic Moment Vector: $\vec{\mu}_n = \gamma \vec{I}$
where γ is called 'gyromagnetic ratio'.
11. $|\vec{\mu}_n| = |\gamma \vec{I}| = \gamma \hbar \sqrt{I(I+1)}$ and $\mu_z = \gamma I_z = \gamma \hbar m_I$
12. Larmor's Frequency: $\omega = \gamma B_o$ where B_o is applied magnetic field.
13. Energy of Interaction: $E = -\mu_z B_o = -\gamma \hbar m_I B_o$
For a proton, $I = \frac{1}{2} \implies m_I = \pm\frac{1}{2}$
 $\implies E_{+1/2} = -\frac{1}{2} \gamma \hbar B_o$, $E_{-1/2} = \frac{1}{2} \gamma \hbar B_o$



15. Selection Rule for NMR: $m_I = \pm 1$.

- (a) Either keep ν constant and keep increasing B_o and when energy matches, the spin flips and gets recorded in the NMR record.
- (b) Or keep B_o constant and keep increase ν .



$$\frac{N_\beta}{N_\alpha} = e^{-\Delta E / K T} = e^{-\gamma \hbar B_o / K T}$$

$$\implies \frac{N_\beta}{N_\alpha} \propto T, \quad \frac{N_\beta}{N_\alpha} \propto \frac{1}{\Delta E}$$

If $T = 300\text{K}$, and $B_o = 1\text{T}$, then population ratio is $0.99993 \approx 1$.

17. The phenomenon that keeps the ratio in equilibrium i.e. population difference remains maintained is called "relaxation".

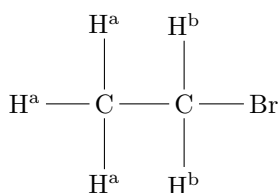
- (a) Spin-Lattice Relaxation: The energy is transferred to the lattice and the spin returns to the ground state.
- (b) Spin-Spin Relaxation: The energy is transferred to the neighboring spins and the spin returns to the ground state.

18. For an induced magnetic field δB , $B_{local} = B_o + \delta B$.

- (a) When $\delta B = (+ve)$, it is called "deshielding field" and energy is less.
- (b) When $\delta B = (-ve)$, it is called "shielding field" and energy is more.

19. Shielded protons resonate at higher magnetic fields.

20. All different kinds of protons resonate at different frequencies.



(a) 'a' type protons have carbon adjacent to Br.

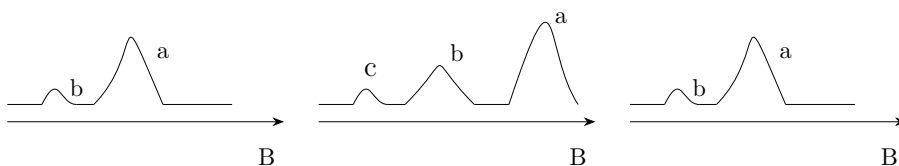
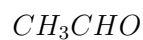
(b) 'b' type protons have carbon adjacent to H.

(c) $\therefore$ 2 types of protons are in the compound.

(d) Hence, 2 NMR signals are expected.

21. Protons next to the most electronegative atom have a lower magnetic field as they have less spin density meaning less shielding.

22. Area under an NMR curve give the number of protons present in given environment.



23.

$$\delta B \propto B_o \implies \delta B = -\sigma B_o \implies B = B_o(1 - \sigma)$$

where σ is the 'shielding constant'.

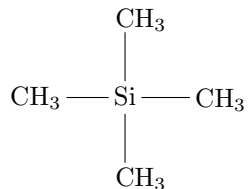
24. If H_A is to be measured with respect to H_B ,

$$B_{H_A} = B_o(1 - \sigma_A) \quad B_{H_B} = B_o(1 - \sigma_B)$$

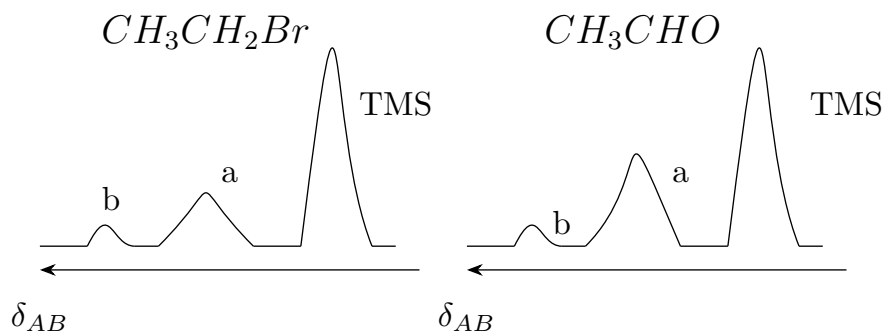
$$B_{H_A} - B_{H_B} = B_o(\sigma_B - \sigma_A) = B_o\delta_{AB} \implies \boxed{\delta_{AB} = \frac{B_{H_A} - B_{H_B}}{B_o} \times 10^6 \text{ ppm}}$$

where δ_{AB} is called the 'chemical shift'.

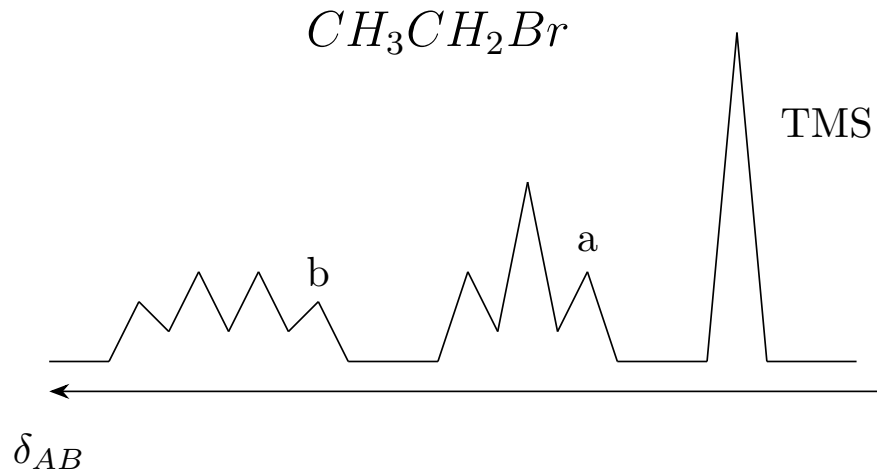
25. **Reference Proton Molecule:** TMS (Tetramethylsilane) is used as a reference molecule.



It has the highest shielded protons. Also it have 12 similar protons, making it have a high intensity. Also it have a very low boiling point.



However, actual NMR diagrams would be:



Number of splitting = $n + 1$, where n is the number of nearby protons.

26. The pascals triangle tells us about the amplitude variance of the splitted proton signals.
27. NMR sample is always measured in deuterated solvent in order for it to

not affect the detection. Example, $CDCl_3$ is used instead of $CHCl_3$ as I value of D is 1, which is not detected in NMR.

4 Phase Rule

4.1 Introduction:

For a system to be in equilibrium, $\Delta G = 0$ with T and P constant.



4.1.1 Gibb's Phase Rule:

It is a general rule for the equilibrium of a system. It is given by the equation:

$$F = C - P + 2$$

where, F = Degrees of freedom, C = Number of components in the system, P = Number of phases in the system.

Phase Rule is only applicable to heterogeneous equilibria.

4.1.2 Phase:

1. A phase is a homogeneous part of a system which is separated from other parts by a physically distinct, mechanically separable, and a clear boundary.
2. Two immiscible liquids are two phases.
3. Two miscible fluids are only one phase.
4. All gases contribute to a single phase.
5. All solids contribute to separate phases.
6. Dissolved solids in liquids are considered to be a singular phase.
7. However for super saturated dissolved solutions, they are considered as multiple phases.
8. For a close container with volatile solids/liquids, a gaseous or vapor state also starts generating which contributes to another phase.

4.1.3 Components:

1. The number of components is the minimum number of constituent participating in equilibrium required to define the number of phases.
Example: $H_2O_{(s)} \rightleftharpoons H_2O_{(l)}$ has $c = 1$ (only H_2O).
2. $c = \text{No. of constituents} - \text{No. of relations} / \text{equilibrium equations}$.
Example: $CaCO_{3(s)} \rightleftharpoons CaO_{(s)} + CO_{2(g)} \implies c = 3 - 1 = 2$.
3. If ions are involved, subtract the electrical neutrality from the value of c.
Example: $NaCl \rightleftharpoons Na^+ + Cl^- \implies c = 3 - 1 - 1 = 1$.

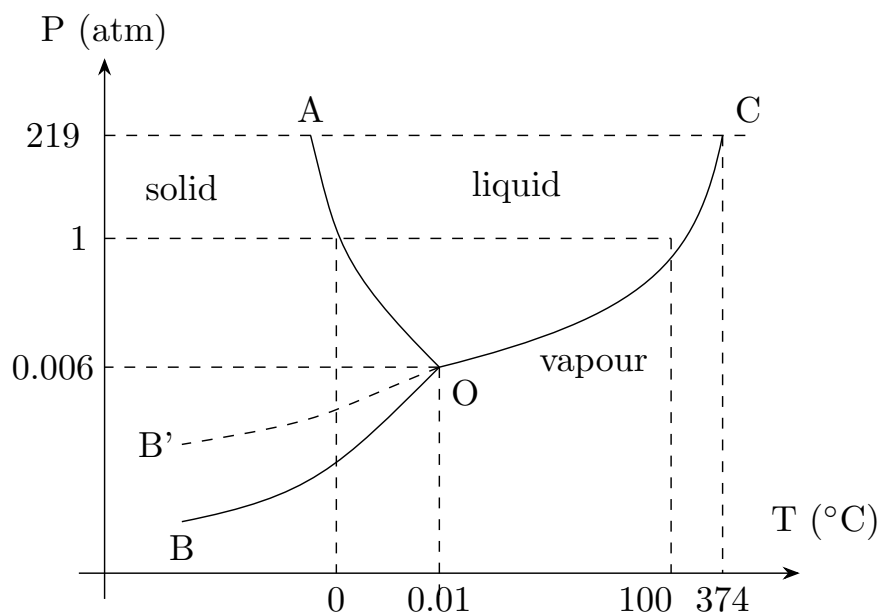
4.1.4 Degrees of Freedom:

1. $H_2O_{(s)} \rightleftharpoons H_2O_{(l)} \implies f = c - p + 2 = 1 - 2 + 2 = 1.$
2. $CaCO_{3(s)} \rightleftharpoons CaO_{(s)} + CO_{2(g)} \implies f = c - p + 2 = 2 - 3 + 2 = 1.$
3. If $f = 0$, the equilibrium is invariant and for only one point it exist.
4. If $f = 1$, it is univariant and change in one variable brings change in another variable for it to exist.

4.1.5 Derivation of Gibb's Phase Rule:

1. $f = \text{No. of Intensive Variables} - \text{No. of relations between them}.$
2. If 'c' components are present in a solution, then 'c-1' variables are required to be known to define the system.
3. If there are 'p' phases, then 'p(c-1)' variables are required to be known to define the system.
4. Pressure and Temperature are two Intensive Variables always present.
5. Hence, No. of Intensive Variables = $p(c-1) + 2.$
6. If 'p' phases are in equilibrium, then 'p-1' relations are between them.
7. If 'c' components are present, then there are 'c(p-1)' relations.
8. Hence, No. of relations between them = $c(p-1).$
9. $f = p(c-1) + 2 - c(p-1) = c - p + 2.$

4.2 Phase Diagram of H₂O:



1. **Curve OA:** $H_2O_{(s)} \rightleftharpoons H_2O_{(l)}$ (Melting / Freezing Curve)

$$c = 1, p = 2 \implies f = 1 - 2 + 2 = 1 \text{ (univariant)}$$

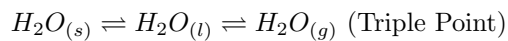
2. **Curve OC:** $H_2O_{(l)} \rightleftharpoons H_2O_{(g)}$ (Boiling / Condensation Curve)

$$c = 1, p = 2 \implies f = 1 - 2 + 2 = 1 \text{ (univariant)}$$

3. **Curve OB:** $H_2O_{(s)} \rightleftharpoons H_2O_{(g)}$ (Sublimation / Deposition Curve)

$$c = 1, p = 2 \implies f = 1 - 2 + 2 = 1 \text{ (univariant)}$$

4. **Point O:** $P = 0.006 \text{ atm}$, $T = 0.01^\circ\text{C}$.



$$c = 1, p = 3 \implies f = 1 - 3 + 2 = 0 \text{ (invariant)}$$

5. **Area AOB:** $H_2O_{(s)}$ (solid region)

$$c = 1, p = 1 \implies f = 1 - 1 + 2 = 2 \text{ (bivariant)}$$

6. **Area AOC:** $H_2O_{(l)}$ (liquid region)

$$c = 1, p = 1 \implies f = 1 - 1 + 2 = 2 \text{ (bivariant)}$$

7. **Area BOC:** $H_2O_{(g)}$ (vapour region)

$$c = 1, p = 1 \implies f = 1 - 1 + 2 = 2 \text{ (bivariant)}$$

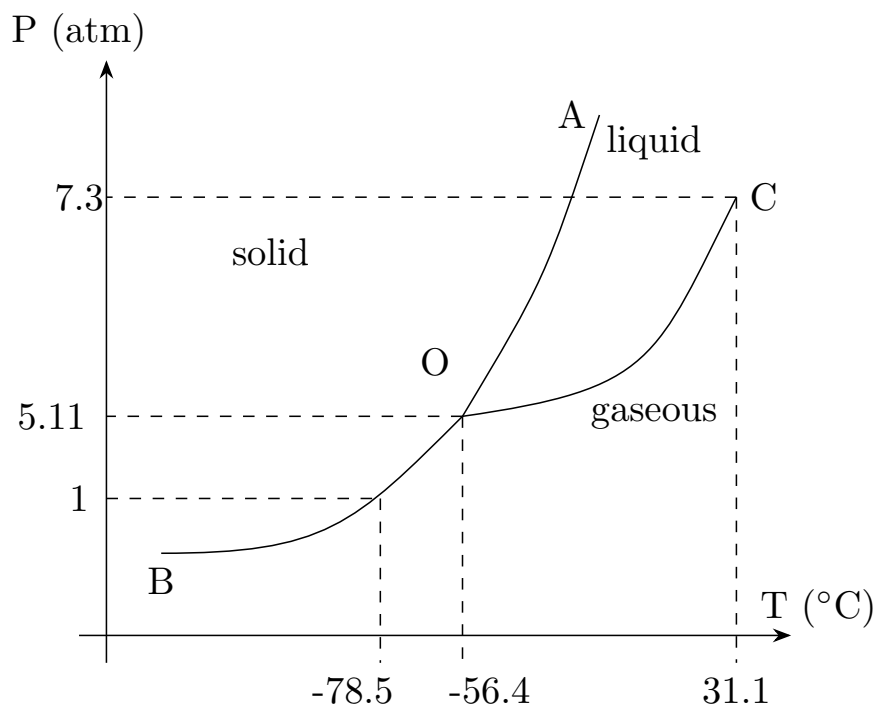
8. **Curve OB':** It is a metastable state for "supercooled liquid".

9. **Supercritical Fluid (SCF):** $P = 219 \text{ atm}$, $T = 374^\circ\text{C}$.

Above these critical parameters, it flows like a liquid, but has density of a gas.

10. On keeping temperature constant and increasing pressure, solid turns into a liquid. This only occurs in water.

4.3 Phase Diagram of CO_2 :



1. **Curve OA:** $\text{CO}_{2(s)} \rightleftharpoons \text{CO}_{2(l)}$ (Melting Curve)

$$c = 1, p = 2 \implies f = 1 - 2 + 2 = 1 \text{ (univariant)}$$

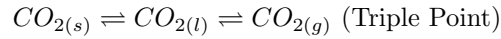
2. **Curve OB:** $CO_{2(s)} \rightleftharpoons CO_{2(g)}$ (Sublimation Curve)

$$c = 1, p = 2 \implies f = 1 - 2 + 2 = 1 \text{ (univariant)}$$

3. **Curve OC:** $CO_{2(l)} \rightleftharpoons CO_{2(g)}$ (Vapourization Curve)

$$c = 1, p = 2 \implies f = 1 - 2 + 2 = 1 \text{ (univariant)}$$

4. **Point O:** $P = 5.11 \text{ atm}$, $T = -56.4^\circ\text{C}$.



$$c = 1, p = 3 \implies f = 1 - 3 + 2 = 0 \text{ (invariant)}$$

5. **Area AOB:** $CO_{2(s)}$ (solid region)

$$c = 1, p = 1 \implies f = 1 - 1 + 2 = 2 \text{ (bivariant)}$$

6. **Area AOC:** $CO_{2(l)}$ (liquid region)

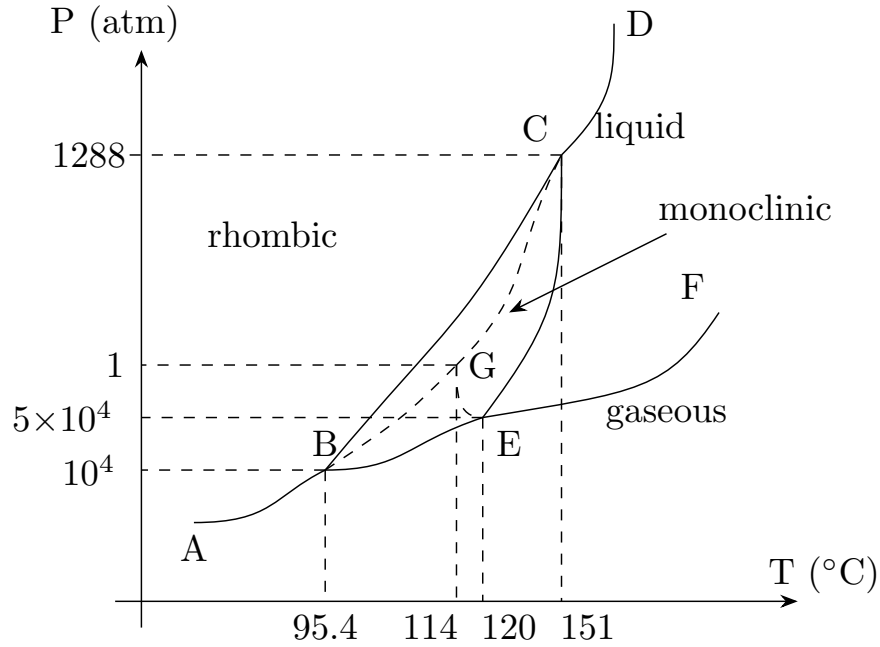
$$c = 1, p = 1 \implies f = 1 - 1 + 2 = 2 \text{ (bivariant)}$$

7. **Area BOC:** $CO_{2(g)}$ (gaseous region)

$$c = 1, p = 1 \implies f = 1 - 1 + 2 = 2 \text{ (bivariant)}$$

8. **Supercritical Fluid:** $P = 7.3 \text{ atm}$, $T = 31.1^\circ\text{C}$.

4.4 Phase Diagram of Sulphur:



1. **Curve AB:** $S_{(rhombic)} \rightleftharpoons S_{(g)}$ (Sublimation Curve)
 $c = 1, p = 2 \implies f = 1 - 2 + 2 = 1$ (univariant)
2. **Curve BE:** $S_{(monoclinic)} \rightleftharpoons S_{(g)}$ (Sublimation Curve)
 $c = 1, p = 2 \implies f = 1 - 2 + 2 = 1$ (univariant)
3. **Curve CD:** $S_{(rhombic)} \rightleftharpoons S_{(l)}$ (Melting Curve)
 $c = 1, p = 2 \implies f = 1 - 2 + 2 = 1$ (univariant)
4. **Curve CE:** $S_{(monoclinic)} \rightleftharpoons S_{(l)}$ (Melting Curve)
 $c = 1, p = 2 \implies f = 1 - 2 + 2 = 1$ (univariant)
5. **Curve EF:** $S_{(l)} \rightleftharpoons S_{(g)}$ (Vapourization Curve)
 $c = 1, p = 2 \implies f = 1 - 2 + 2 = 1$ (univariant)
6. **Curve BC:** $S_{(rhombic)} \rightleftharpoons S_{(monoclinic)}$ (Transition Curve)
 $c = 1, p = 2 \implies f = 1 - 2 + 2 = 1$ (univariant)

7. **Point B:** $P = 10^{-4}$ atm, $T = 95.4^\circ\text{C}$.

$$S_{(rhombic)} \rightleftharpoons S_{(monoclinic)} \rightleftharpoons S_{(g)} \text{ (Triple Point)}$$

$$c = 1, p = 3 \implies f = 1 - 3 + 2 = 0 \text{ (invariant)}$$

8. **Point C:** $P = 1288$ atm, $T = 151^\circ\text{C}$.

$$S_{(rhombic)} \rightleftharpoons S_{(monoclinic)} \rightleftharpoons S_{(l)} \text{ (Triple Point)}$$

$$c = 1, p = 3 \implies f = 1 - 3 + 2 = 0 \text{ (invariant)}$$

9. **Point E:** $P = 5 \times 10^{-4}$ atm, $T = 120^\circ\text{C}$.

$$S_{(monoclinic)} \rightleftharpoons S_{(l)} \rightleftharpoons S_{(g)} \text{ (Triple Point)}$$

$$c = 1, p = 3 \implies f = 1 - 3 + 2 = 0 \text{ (invariant)}$$

10. **Point G:** $P = 1$ atm, $T = 114^\circ\text{C}$.

$$S_{(rhombic)} \rightleftharpoons S_{(l)} \rightleftharpoons S_{(g)} \text{ (Metastable Triple Point)}$$

$$c = 1, p = 3 \implies f = 1 - 3 + 2 = 0 \text{ (invariant)}$$

11. **Area ABCD:** $S_{(rhombic)}$ (Rhombic Region)

$$c = 1, p = 1 \implies f = 1 - 1 + 2 = 2 \text{ (bivariant)}$$

12. **Area BCE:** $S_{(monoclinic)}$ (Monoclinic Region)

$$c = 1, p = 1 \implies f = 1 - 1 + 2 = 2 \text{ (bivariant)}$$

13. **Area DCEF:** $S_{(l)}$ (Liquid Region)

$$c = 1, p = 1 \implies f = 1 - 1 + 2 = 2 \text{ (bivariant)}$$

14. **Area ABEF:** $S_{(g)}$ (Gaseous Region)

$$c = 1, p = 1 \implies f = 1 - 1 + 2 = 2 \text{ (bivariant)}$$

4.5 Two Components Systems:

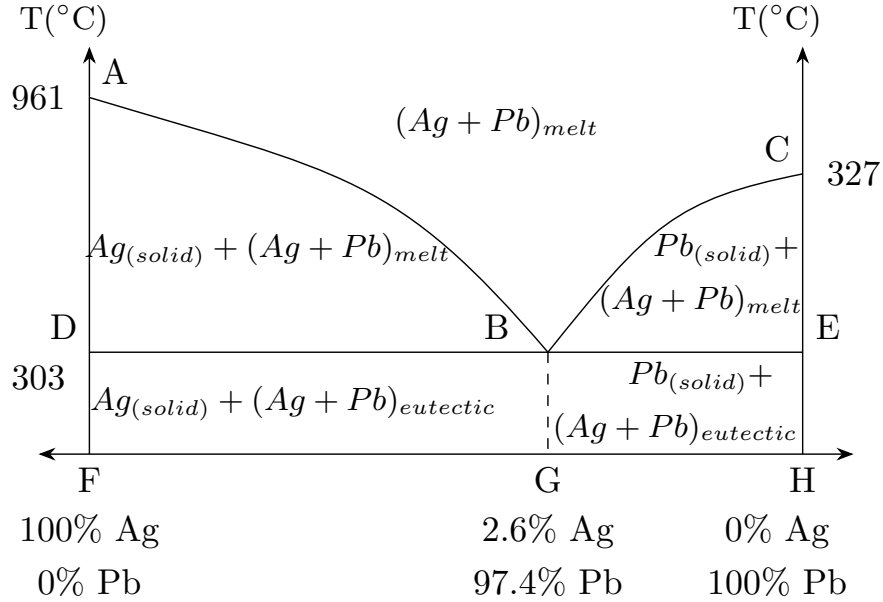
1. **Reduced Phase Rule:** $f = c - p + 1$.

2. Here the Pressure term is constant at 1 atm.

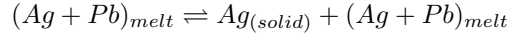
3. For alloy systems, vapour phase is negligible.

4. **Eutectic Mixture:** A constant melting mixture.

4.6 Phase Diagram of Ag/Pb System:

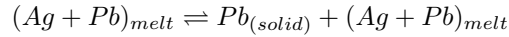


1. **Curve AB:** Freezing / Melting Curve of Ag.



$$c = 2, p = 2 \implies f' = 2 - 2 + 1 = 1 \text{ (univariant)}$$

2. **Curve BC:** Freezing / Melting Curve of Pb.



$$c = 2, p = 2 \implies f' = 2 - 2 + 1 = 1 \text{ (univariant)}$$

3. **Point B:** $(Ag + Pb)_{melt} \rightleftharpoons (Ag + Pb)_{eutectic}$ (Eutectic Point)

$$c = 2, p = 3 \implies f' = 2 - 3 + 1 = 0 \text{ (invariant)}$$

4. **Area ABD:** $Ag_{(solid)} + (Ag + Pb)_{melt}$

$$c = 2, p = 2 \implies f' = 2 - 2 + 1 = 1 \text{ (univariant)}$$

5. **Area CBE:** $Pb_{(solid)} + (Ag + Pb)_{melt}$

$$c = 2, p = 2 \implies f' = 2 - 2 + 1 = 1 \text{ (univariant)}$$

6. **Area DBGF:** $Ag_{(solid)} + (Ag + Pb)_{eutectic}$

$$c = 2, p = 2 \implies f' = 2 - 2 + 1 = 1 \text{ (univariant)}$$

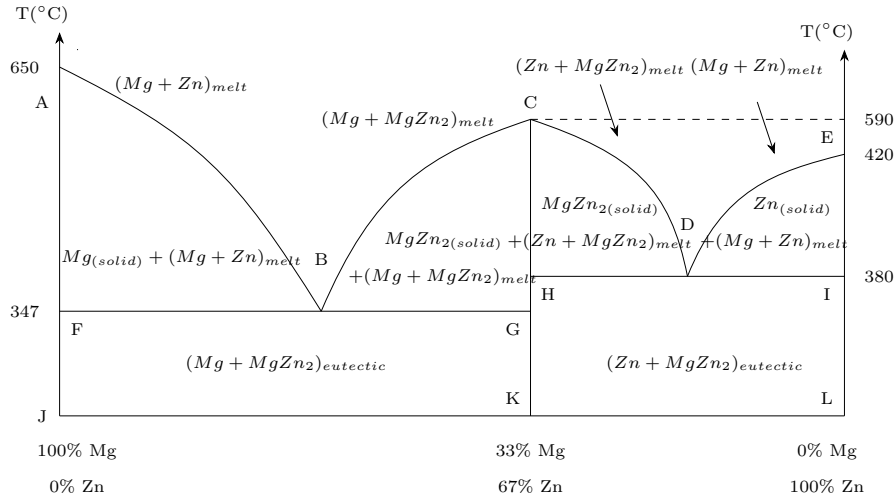
7. **Area BEHG:** $Pb_{(solid)} + (Ag + Pb)_{eutectic}$

$$c = 2, p = 2 \implies f' = 2 - 2 + 1 = 1 \text{ (univariant)}$$

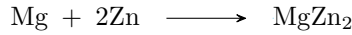
8. **Area ABC:** $(Ag + Pb)_{melt}$

$$c = 2, p = 1 \implies f' = 2 - 1 + 1 = 2 \text{ (bivariant)}$$

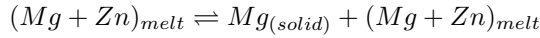
4.7 Phase Diagram of Mg/Zn System:



Here a reaction occur at a 1:2 Ratio of Mg:Zn.

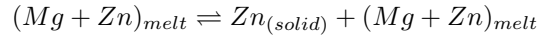


1. **Curve AB:** Freezing / Melting Curve of Mg.



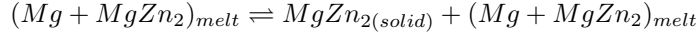
$$c = 2, p = 2 \implies f' = 2 - 2 + 1 = 1 \text{ (univariant)}$$

2. **Curve DE:** Freezing / Melting Curve of Zn.



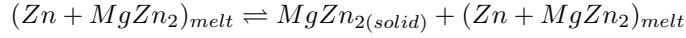
$$c = 2, p = 2 \implies f' = 2 - 2 + 1 = 1 \text{ (univariant)}$$

3. **Curve BC:** Freezing / Melting Curve of $MgZn_2$ on addition of Mg.



$$c = 2, p = 2 \implies f' = 2 - 2 + 1 = 1 \text{ (univariant)}$$

4. **Curve CD:** Freezing / Melting Curve of $MgZn_2$ on addition of Zn.



$$c = 2, p = 2 \implies f' = 2 - 2 + 1 = 1 \text{ (univariant)}$$

5. **Area ABF:** $Mg_{(solid)} + (Mg + Zn)_{melt}$

$$c = 2, p = 2 \implies f' = 2 - 2 + 1 = 1 \text{ (univariant)}$$

6. **Area DIE:** $Zn_{(solid)} + (Mg + Zn)_{melt}$

$$c = 2, p = 2 \implies f' = 2 - 2 + 1 = 1 \text{ (univariant)}$$

7. **Area BCG:** $MgZn_{2(solid)} + (Mg + MgZn_2)_{melt}$

$$c = 2, p = 2 \implies f' = 2 - 2 + 1 = 1 \text{ (univariant)}$$

8. **Area CHD:** $MgZn_{2(solid)} + (Zn + MgZn_2)_{melt}$

$$c = 2, p = 2 \implies f' = 2 - 2 + 1 = 1 \text{ (univariant)}$$

9. **Area FGKJ:** $(Mg + MgZn_2)_{eutectic}$

$$c = 2, p = 2 \implies f' = 2 - 2 + 1 = 1 \text{ (univariant)}$$

10. **Area HILK:** $(Zn + MgZn_2)_{eutectic}$

$$c = 2, p = 2 \implies f' = 2 - 2 + 1 = 1 \text{ (univariant)}$$

11. **Point B:** $(Mg + MgZn_2)_{melt} \rightleftharpoons (Mg + MgZn_2)_{eutectic}$ (Eutectic Point)

$$c = 2, p = 3 \implies f' = 2 - 3 + 1 = 0 \text{ (invariant)}$$

12. **Point D:** $(Zn + MgZn_2)_{melt} \rightleftharpoons (Zn + MgZn_2)_{eutectic}$ (Eutectic Point)

$$c = 2, p = 3 \implies f' = 2 - 3 + 1 = 0 \text{ (invariant)}$$

13. **Point C:** $MgZn_{2(melt)} \rightleftharpoons MgZn_{2(eutectic)}$ (Congruent Melting Point)

$$c = 2, p = 3 \implies f' = 2 - 3 + 1 = 0 \text{ (invariant)}$$

14. **Area Above AB:** $(Mg + Zn)_{melt}$

$$c = 2, p = 1 \implies f' = 2 - 1 + 1 = 2 \text{ (bivariant)}$$

15. **Area Above DE:** $(Mg + Zn)_{melt}$

$$c = 2, p = 1 \implies f' = 2 - 1 + 1 = 2 \text{ (bivariant)}$$

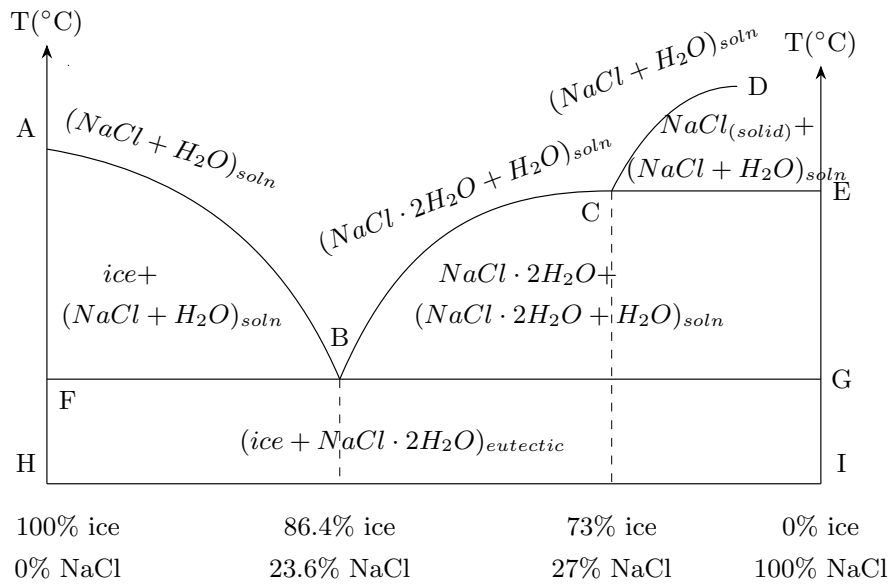
16. **Area Above BC:** $(Mg + MgZn_2)_{melt}$

$$c = 2, p = 1 \implies f' = 2 - 1 + 1 = 2 \text{ (bivariant)}$$

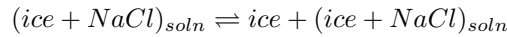
17. **Area Above CD:** $(Zn + MgZn_2)_{melt}$

$$c = 2, p = 1 \implies f' = 2 - 1 + 1 = 2 \text{ (bivariant)}$$

4.8 Phase Diagram of NaCl/H₂O System:

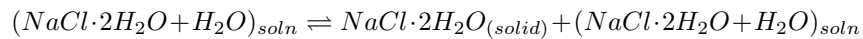


1. **Curve AB:** Freezing / Melting Curve of H₂O.



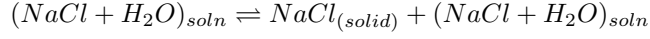
$$c = 2, p = 2 \implies f' = 2 - 2 + 1 = 1 \text{ (univariant)}$$

2. **Curve BC:** Solubility Curve of NaCl · 2H₂O.



$$c = 2, p = 2 \implies f' = 2 - 2 + 1 = 1 \text{ (univariant)}$$

3. **Curve CD:** Solubility Curve of NaCl.



$$c = 2, p = 2 \implies f' = 2 - 2 + 1 = 1 \text{ (univariant)}$$

4. **Area ABF:** $ice + (ice + NaCl)_{soln}$

$$c = 2, p = 2 \implies f' = 2 - 2 + 1 = 1 \text{ (univariant)}$$

5. **Area BCEG:** $NaCl \cdot 2H_2O_{(solid)} + (NaCl \cdot 2H_2O + H_2O)_{soln}$

$$c = 2, p = 2 \implies f' = 2 - 2 + 1 = 1 \text{ (univariant)}$$

6. **Area CDE:** $NaCl_{(solid)} + (NaCl + H_2O)_{soln}$

$$c = 2, p = 2 \implies f' = 2 - 2 + 1 = 1 \text{ (univariant)}$$

7. **Area FGIH:** $(ice + NaCl \cdot 2H_2O)_{eutectic}$

$$c = 2, p = 2 \implies f' = 2 - 2 + 1 = 1 \text{ (univariant)}$$

8. **Area Above AB:** $(NaCl + H_2O)_{soln}$

$$c = 2, p = 1 \implies f' = 2 - 1 + 1 = 2 \text{ (bivariant)}$$

9. **Area Above BC:** $(NaCl \cdot 2H_2O + H_2O)_{soln}$

$$c = 2, p = 1 \implies f' = 2 - 1 + 1 = 2 \text{ (bivariant)}$$

10. **Area Above CD:** $(NaCl + H_2O)_{soln}$

$$c = 2, p = 1 \implies f' = 2 - 1 + 1 = 2 \text{ (bivariant)}$$

11. **Point B:** $(H_2O + NaCl \cdot 2H_2O)_{soln} \rightleftharpoons (ice + NaCl \cdot 2H_2O)_{eutectic}$

$$c = 2, p = 3 \implies f' = 2 - 3 + 1 = 0 \text{ (invariant)}$$

12. **Point C:** $(H_2O + NaCl)_{soln} \rightleftharpoons NaCl_{(solid)} + H_2O_{(l)}$

$$c = 2, p = 3 \implies f' = 2 - 3 + 1 = 0 \text{ (invariant)}$$

The system decomposes before reaching a sharp melting point.

4.9 Three Component System:

For three component system, both temperature and pressure are constant.

$$f'' = c - p$$

